Team: Cyber Guardians

Project: Secure Student WIL Hours Tracker

Assessment Framework: OWASP ASVS Level 1

Assessment Date: November 2026

Security Status: Conditionally Approved for Development Phase

1. Executive Summary

The purpose of the Security Posture Baseline Assessment is to evaluate the proposed Secure Student WIL Hours Tracker against a recognized security baseline before development and deployment. The OWASP Application Security Verification Standard (ASVS) Level 1 assessment will be used as a foundational security framework.

The proposed project will process sensitive information including student details, attendance records, uploaded evidence, mentor approvals and generate reports. To protect the confidentiality, integrity and availability of the project information security controls must be implemented.

The OWASP Application Security Verification Standard (ASVS) Level 1 assessment helps identify the current security considerations, existing security gaps and recommended controls that should be implemented throughout the project lifecycle.

2. Project Description

The Secure Student WIL Hours Tracker is a cloud-based application that is designed to assist students in recording, monitoring and verifying the 320 Workplace Integrated Learning (WIL) hours that are required.

The system will do the following:

Student login and authentication

Clock-in and clock-out with time synchronization

Photo evidence uploads

Hours tracking dashboard

Mentor approval workflow

Administrative management features

The application will be deployed using Microsoft Azure cloud services and will be accessed by students, mentors, coordinators and administrators.

3. Assessment Scope

The following project components are included in this baseline assessment:

Application Components:

User authentication and User authorization

Attendance recording and Session management

Evidence uploads

Cloud Components:

Azure App Service

Azure SQL Database

Azure Blob Storage

Microsoft Entra ID

Azure Key Vault

Information Assets:

Student and Mentor information

Attendance records

Uploaded evidence

Audit logs

Project reports

4. Baseline Security Assessment

Security Control	Status	Priority	####
User authentication	Planned	High	Microsoft Entra ID implemented
Multi-factor authentication	Planned	High	Enabled on all users
Role-Based Access control	Planned	High	The principle of least privilege must be followed.
Session management	Planned	High	Session timeout controls must be implemented.
Secure file upload controls	Planned	High	Uploaded evidence must never be tampered with.
Strong password policy	Planned	High	A strong password with complex characters is required for all users.
Audit logging	Planned	High	All user actions must be logged.
POPIA Compliance controls	Planned	High	Personal information requires protection.
Access review process	Planned	High	User permission and privileges must always be monitored.
Device security requirements	Planned	High	Endpoint protection must be enabled on all users.
Backup and Disaster recovery procedures	Planned	High	Backup plans must be implemented alongside disaster recovery plans.

5. Security Readiness Assessment

The proposed Secure Student WIL Hours Tracker project is currently in the planning and requirements phase, with most security controls been already identified and incorporated into the proposed project design, but they have not yet been implemented. The project's security posture can therefore be classified as Planned but Not Operational.

6. Identified Security Risks

Risk	Impact	Likelihood
Weak user passwords	High	Medium
Data exposure due to poor access controls.	High	Low
Unauthorized access to student records.	High	Medium
Students modifying attendance records.	High	Medium
Misuse of privileged accounts.	High	Low
Account compromise through phishing.	High	Medium
Failure to meet POPIA compliance requirements.	High	Low
Loss of uploaded evidence	Medium	Low

7. Security Gaps Identified

High-Priority Gaps

The following security controls have not yet been implemented and should be prioritized for the identified security risks.

Multi-Factor authentication

Role-Based access control

Encryption of sensitive information

Secure file upload validation

Audit logging

Secure session management

Medium Priority Gaps

Continuous security monitoring

Backup and recovery procedures

Incident response process

User access review processes

8. OWASP ASVS Level 1 Review

The following OWASP ASVS areas were reviewed during the baseline assessment:

V2 Authentication

V3 Session Management

V4 Access Control

V8 Data Protection

V10 Malicious Input Handling

V14 Configuration Security

9. Recommended Security Controls

The following security controls should be implemented throughout development of the proposed project Secure Student WIL Hours Tracker:

Authentication Controls

Microsoft Entra ID integration

Multi-Factor Authentication

Strong password requirements

Account lockout controls

Access Control Controls

Role-Based Access Control

Least Privilege Principle

Administrative access restrictions

Data Protection Controls

Encryption in transit using HTTPS/TLS

Encryption at rest for stored data

Secure storage of uploaded evidence

Azure Key Vault for secrets management

Monitoring Controls

Audit logging and user activity monitoring

Security event alerts

Log retention procedures

Compliance Controls

POPIA compliance requirements

User consent mechanisms

Data retention procedures

Privacy protection controls

10. NIST RMF Alignment

The assessment aligns with the NIST Risk Management Framework principles:

Identify

Assets identified:

Student information

Attendance records

Uploaded evidence

Mentor approvals

Protect

Proposed controls:

MFA

RBAC

Encryption

Secure cloud services

Detect

Planned controls:

Audit logging

Monitoring

Security event tracking

Recover

Future requirement:

Azure backup and recovery strategy

11. Conclusion

The proposed Secure Student WIL Hours Tracker demonstrates a strong foundation for a secure development by incorporating security considerations at the planning stage. The project has identified critical security controls such as Microsoft Entra ID authentication, Multi-Factor Authentication, Role-Based Access Control, audit logging, encryption and POPIA compliance requirements.

While most controls remain in the planning phase, the project has successfully established a security baseline aligned with OWASP ASVS Level 1 and NIST Risk Management Framework principles. Future development activities should focus on implementing, testing and validating these controls to ensure the confidentiality, integrity and availability of project information.